Homework #2

Bayes Filters

BF-Q1

$$Bel(x_t) = p(x_t \mid u_1, z_1, \cdots u_t, z_t) \quad (1)$$

$$Bel(x_t) = \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1})\, Bel(x_{t-1})\, dx_{t-1} \quad (2)$$

label correction step, update step & prior term in (2)
& derive 2 from 1, state rule/law/assumption used

$$Bel(x_t) = \underbrace{\eta\, p(z_t \mid x_t)}_{\text{correction step}} \int \underbrace{p(x_t \mid u_t, x_{t-1})\, \underbrace{Bel(x_{t-1})}_{\text{prior term}}\, dx_{t-1}}_{\text{update step}}$$

$$Bel(x_t) = p(x_t \mid u_1, z_1, \cdots u_t, z_t)$$

① bayes rule $= \eta\, p(z_t \mid x_t)\, p(x_t \mid u_1, z_1, \cdots u_t)$

② markov assum. $= \eta\, p(z_t \mid x_t)\, p(x_t \mid u_1, z_1, \cdots, u_t)$

③ law of total prob $= \eta\, p(z_t \mid x_t) \int p(x_t \mid u_1, z_1, \cdots u_t, x_{t-1})\, p(x_{t-1} \mid u_1, z_1, \cdots u_t)\, dx_{t-1}$

④ markov assum. $= \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1})\, p(x_{t-1} \mid u_1, z_1, \cdots u_t)\, dx_{t-1}$

⑤ markov assum $= \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1})\, p(x_{t-1} \mid u_1, z_1, \cdots z_{t-1})\, dx_{t-1}$

⑥ $Bel(x_t) = \eta\, p(z_t \mid x_t) \int p(x_t \mid u_t, x_{t-1})\, Bel(x_{t-1})\, dx_{t-1}$

~~occupancy grid mapping~~

BF-Q2) A markov assumption states that a variable is only dependant on some variables immediately preceding it and not all prior variables

BF-Q3) $p(x_{t+1}=h \mid u_{t+1}=r, x_t=\neg h) = 0.7$

$p(\neg h) = 0.7$

$P_1(x_1 \mid u_1=r, x_0) = \sum P(h \mid u, x_0) P(x_0 \mid z_1)$

law of total probability

$= P(h \mid u, \neg h) P(\neg h) + P(h \mid u, h) P(h)$

$= (0.7)(0.7) + (0.3)(0.3)$

$= 0.49 + 0.3$

$= 0.79$ (1st try)

$P_2(x_2=h \mid u_2=r) = P(h_2 \mid u_2, \neg h_1) P(\neg h_1) + P(h_2 \mid u_2, h_1) P(h_1)$

$= (0.7)(0.21) + (1)(0.79)$

$= (0.147) + 0.79$

$= 0.937$ 2nd try

BF-Q4) $P(x_{t+1}=c \mid x_t=b, u_{t+1}=p) = 0.9$

$P(z=c \mid x=b) = 0.2$

$P(z=c \mid x=c) = 0.7$

$u_{t+1} = p$; $z_{t+1} = c$

$P(x_{t+1}=b \mid z_{t+1}=c, u_t=p)$ ->?

uniform prior ($P(x_0=b) = P(x_0=c) = 0.5$)

$Bel(x_{t+1}) = \eta\, p(z_t \mid x_t) \sum p(x_t \mid u_t, x_{t-1}) Bel(x_{t-1})$

$P(x_t=c) = \eta\, p(z_t \mid x_t) \left(P(x_t=c \mid u_t, x_{t-1}) Bel(x_{t-1}) + P(x_t=b \mid u_t, x_{t-1}) Bel(x_{t-1}) \right)$

$= \eta\, p(z_t \mid x_t) \left((0.9)(0.5) + (1)(0.5) \right)$

$= \eta\, p(z_t \mid x_t)(0.95)$

$\eta = \frac{1}{\sum P(z=c(x_i) P(x_i)} = \frac{1}{(0.2)(0.05) + (0.7)(0.95)}$

$0.01 + 0.665$

$\eta = \frac{1}{0.675}$

$= \left(\frac{1}{0.675}\right)(0.7)(0.95)$

$P(x_t=c) = 0.985$ $P(x_t=b) = 1 - P(x_t=c) = 0.015$

~~OGM~~

OGM

OGM-Q1) major assumptions:

① Area tied to a cell is totally free or occupied
② World is static
③ cells are indep of each other

OGM-Q2) log odds is good because updates are additive & is ~~close to 0 or 1~~ good for problems close to 0 or 1

OGM-Q3) ① measurement recieved
② treat cells passed through before measurement as "free" + cells that could be in range of measurement as "occupied", and other cells as "prior"
③ use these in the update step of log odds

OGM-Q4) This alg can fail if we have conflicting measurements from previous or multiple scans, or if our OG assumptions are not true

330
250
210

Expectation & Covariance

EC-Q1) $E[x+y]$ is the expected value with two variables or $E[x+y] = E[x] + E[y]$

$$E[x+y] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)\,dx\,dy$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left(xf(x,y) + yf(x,y)\right)dx\,dy$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)\,dx\,dy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)\,dx\,dy$$
$$= \int x\left(\int f(x,y)\,dy\right)dx + \int y\left(\int f(x,y)\,dx\right)dy$$
$$= \int x f(x)\,dx + \int y f(y)\,dy$$
$$= E[x] + E[y]$$

EC-Q2) a) $\begin{bmatrix}5 & 3\\ 3 & 9\end{bmatrix}$ - valid

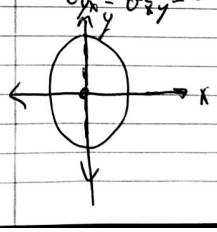
b) $\begin{bmatrix}4 & -8\\ -8 & 9\end{bmatrix}$ - invalid ∴ negative determinant

c) $\begin{bmatrix}2 & 0\\ 6 & -8\end{bmatrix}$ - invalid ∴ negative on diagonal

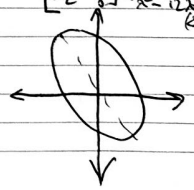
d) $\begin{bmatrix}3 & 2\\ 2 & 10\end{bmatrix}$ - valid

e) $\begin{bmatrix}6 & 1\\ 2 & 10\end{bmatrix}$ - invalid (assymetric

EC-Q3) a) $\begin{bmatrix}4 & 0\\ 0 & 8\end{bmatrix}$ $\sigma_{xx}^2 = 4$ $\sigma_{yy}^2 = 8$ $\sigma_{xy}^2 = \sigma_{yx}^2 = 0$



b) $\begin{bmatrix}4 & -2\\ -2 & 8\end{bmatrix}$ $\sigma_{xx} = 2$ $\sigma_{yy} = 2\sqrt{2}$



c) $\begin{bmatrix}4 & 3\\ 3 & 4\end{bmatrix}$

